Assisting Stroke Gait with Cable Driven Lower Limb Rehabilitation Exoskeleton (C-LREX): Simulation Study

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ABSTRACT

Cable driven exoskeletons have demonstrated promising results in the rehabilitation of stroke survivors. Usually, the performance of an exoskeleton is tested via multiple experimental trials to allow iterative adjustment and optimization, which can be costly and time-consuming. Preliminary numerical model studies which quantify the effects of various design parameters on the exoskeleton's behavior and compare the responses of the impaired limb with and without the exoskeleton are valuable as they provide time and cost-effective means to fine-tune key design parameters while minimizing unnecessary testing. This study employed a generalized framework of cable driven exoskeleton analysis to investigate the ability of a cable driven lower limb rehabilitation exoskeleton (C-LREX) to correct stroke impaired gait and track reference healthy trajectories. Five different impaired gait patterns were studied. The simulation results demonstrated that the C-LREX model was able to assist the impaired limb to track the reference trajectory successfully in all impaired gait patterns except in the case of high deviation in the range of motion, which required a cable tension that exceed the allowable limit implemented in the model. This study confirmed that our developed simulation framework was able to test the C-LREX model ability to assist various impaired lower limb patterns in the sagittal plane during gait and determine the cable tension requirements. Future work includes expanding the framework to incorporate the frontal plane motion toward testing the performance of C-LREX in assisting three-dimensional impaired motion.

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CCS CONCEPTS •Mathematics of computing → Mathematical analysis → Mathematical optimization •Human-centered computing → Human computer interaction (HCI) •General and reference → Cross-computing tools and techniques → Design •Human-centered computing → Interaction design → Systems and tools for interaction design

KEYWORDS

Exoskeleton, lower limb rehabilitation, cable-driven exoskeleton, link-based model, cable routing, torque mapping

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1 INTRODUCTION

Stroke usually results in movement impairment, and hemiparesis is one of the most commonly observed difficulties [1]. The impairment results in an alteration of the lower limb gait cycle such as a reduced range of motion of lower limb joints (ankle, knee, and hip) [2], muscle spasticity, and abnormal muscle activation patterns, to name a few. In literature, numerous types of exoskeletons have been developed and validated for the rehabilitation of stroke, including link driven [3, 4], and cable driven [5, 6] systems. Cable driven exoskeletons are preferred over link-driven exoskeletons for stroke rehabilitation due to numerous benefits, such as lightweight design, easier reconfigurability, reduced donning off/on time, and transparent design [7].

In literature, the lower limb assisted by an exoskeleton is typically modeled as rigid links incorporating various inertial parameters. Cable driven exoskeletons employ the rigid link-based lower limb model and apply joint moments via cable tension. When tested on stroke patients, cable driven exoskeletons demonstrated promising results in rehabilitation [8] by assisting the subject to track prescribed trajectories. However, the fabrication of an exoskeleton

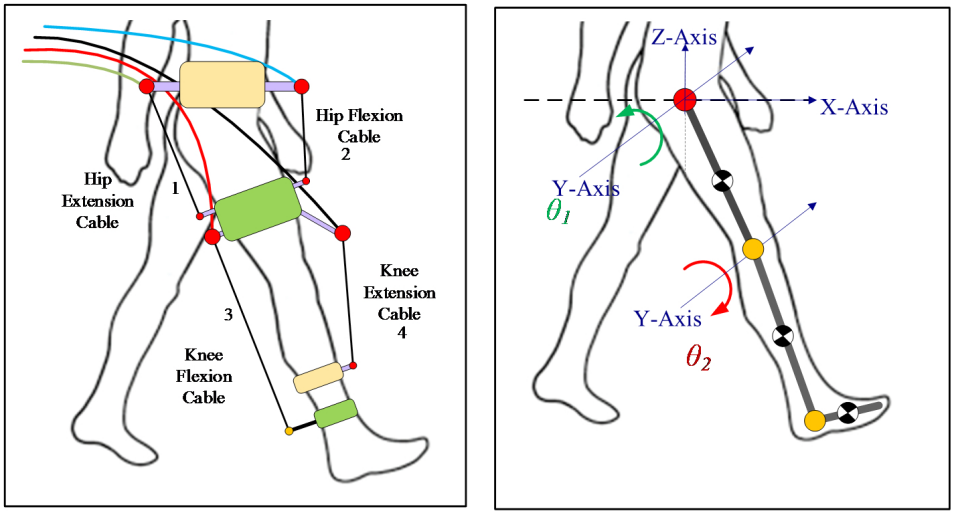

Figure 1: Four Cable-driven conceptual model (left) and a lower limb model for C-LREX (right) [10]

generally involves multiple iterative adjustments and experimental tests on human subjects which tends to be costly and time-consuming. Model-based simulation studies, on the other hand, provide parametric tools to identify the key design parameters and thus save time and effort prior to actual experimentation. Furthermore, such studies are important as they provide insight into the experimental aspect and help determine the requirements during the experiment, leading towards a somehow optimized experiment.

In our previous works [9–11], we developed a pendulum-based model of the lower limb, incorporated cables to generate the motion, and proposed a generalized framework for the analysis of a cable driven lower limb exoskeleton (both in the sagittal plane and combined sagittal-frontal plane). The framework assumed no active contribution of the lower limb but accounted for moments generated by passive tissues. The framework allowed variation in the number of cables, routings, cuff locations, subject anthropometric data, reference trajectory, etc. to analyze the influence of these parameters on tracking performance and design requirements. We generated several routing-based designs of C-LREX and employed the framework to identify a suitable configuration. Furthermore, the framework also predicted the required cable tension and the motor/actuator specification to meet the tracking performance and estimated the optimal cuff parameters.

In this work, we employed the generalized framework to study the ability of the C-LREX to correct the trajectories of five different stroke gait patterns in the sagittal plane [12] and to determine the required cable tension. The lower limb model was personalized using the anthropometry of one stroke patient [13]. The suggested simulation-based framework allows studying the ability and determine the limitations of a given configuration of cables (number, routing, etc.) and cuff locations in C-LREX to correct impaired gait without the need of prototyping the exoskeleton.

2 METHODOLOGY

In our recent works [9, 10], we developed a generalized framework for the analysis of cable driven rehabilitation exoskeleton (C-LREX) which allowed the determination of suitable cable routing configurations that can be used in the design of C-LREX. Among the studied routings, a 4-cable routing (Figure 1) provided the best overall performance in terms of gait trajectory tracking in the sagittal plane along with model requirements. Furthermore, the 4-cable design has demonstrated its capability to generate the joint moment at the hip and knee joint independently as it uses a pair of cables employed independently for each joint.

The framework models the lower limb as a two-link pendulum with two degrees of freedom (DOF); hip flexion/extension and knee flexion/extension. The foot is assumed to be fixed perpendicularly to the shank. The equation of motion is derived as [10]:

$$M(\theta)\,\ddot{\theta} + C\left(\theta, \dot{\theta}\right)\dot{\theta} + G(\theta) = \tau \tag{1}$$

Where *M, C, and G* are the inertial matrix, Coriolis component, and Gravitational components respectively and τ represents the torques on the joints.

A proportional-derivative (PD) controller based 3 layer control architecture was adopted to track the reference healthy trajectory [10] as shown in Figure 2. The upper control layer employs a PD controller to generate the required torque based on the tracking error. The state estimation layer estimates various data required in other control layers such as *M, C, and G,* as well as the subject contribution. The lower control layer distributes the desired torque among the cables as cable tension via a quadratic programming approach. In previous work, the passive elastic joint moment [14] (due to passive tissues around the joint) was considered the only contribution of the subject. To accommodate the stroke survivor's actual contribution, the equivalent joint moment (active contribution) corresponding to stroke kinematics is estimated via equation 1).

$$\tau_s = M(\theta_s)\,\ddot{\theta}_s + C\left(\theta_s, \dot{\theta}_s\right)\dot{\theta}_s + G(\theta_s) \tag{2}$$

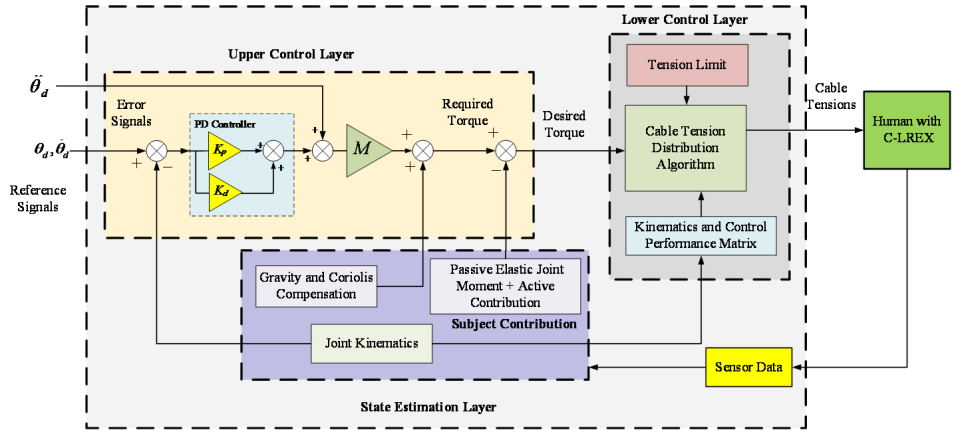

Figure 2: Three-layer control architecture [10]

Table 1: Anthropometric data measurement

Parameter	Value	Unit	Parameter	Value	Unit
m1	7.58	kg	I1	0.1527	kg.m2
m2	3.52	kg	I2	0.0606	kg.m2
m3	1.1	kg	I3	0.0075	kg.m2
L1	0.44	m	a1	0.1902	m
L2	0.43	m	a2	0.1879	m
L3	0.17	m	a3	0.0870	m

The total contribution of the subject during trajectory tracking is assumed to be the sum of the passive elastic joint moment and the moment estimated using the rigid link model (equation 2)).

$$\tau_{s_stroke} = \tau_{passive} + \tau_s \tag{3}$$

2.1 Anthropometry and Trajectory

A reference healthy gait trajectory with gait time of 3.48 seconds was adopted from the dataset published by Fukuchi et al. [13]. The slower gait time was adopted to resemble the slow walking speed of stroke survivors. Kinematics of five different impaired gait patterns of stroke survivors, referred as stroke gait 1-5, were taken from clustered gait data available in the literature [12]. The stroke gait data was collected from 36 stroke patients over a period of six months, during which six different clusters were identified via simultaneously performing clustering and classification from 111 gait cycles. The level of impairment (deviation from the healthy reference) is different in each gait cycle. The trajectory time for stroke survivors' gait was also assumed to be the same as the healthy trajectory time for simulation. The subject anthropometric data was estimated via the model by Winter [15] and is listed in Table I where *I, m, L, and a* represent the moment of inertia, mass, length of the link, and distance of CG of the link from the joint center, respectively. Subscripts *1, 2, and 3* of the parameters refer to thigh, shank, and foot, respectively.

In the current study, it was also assumed that the impaired leg can achieve the workspace (range of motion) of a healthy leg (ignoring the muscle spasticity effect). Based on the stroke subject contribution, the C-LREX exerts the joint moments via the cable tension to force the limb to track the reference healthy trajectory. The leg was assumed suspended in the air (ignoring the stance phase of the motion) since the model was intended only to assist during the swing phase of the motion. The cable tension range was limited to 7-100 N to ensure the cable is always taut and stays within a limit.

3 SIMULATION RESULT

The model simulated one gait cycle and the tracked trajectories, cable tension, and ankle position tracking were analyzed.

3.1 Tracking with Stroke Gait 1

The model successfully forced the impaired gait to follow the healthy reference gait closely. The joint angle tracking performance for stroke gait 1 is illustrated in Figure 3. The cable tension remained within the specified ranges (Figure 4). The ankle position was measured by taking the hip joint as a reference point. The stroke, corrected (model tracked), and reference trajectories were compared (Figure 4).

The initial deviations in the hip and knee joint angles of the model tracked (corrected) trajectory from the healthy trajectory are due to C-LREX starting boundary condition where the model starts from the initial position of the stroke gait (altered ROM of the hip and knee) and were minimized quickly (after < 10% of gait cycle) by C-LREX during walking. Furthermore, in our previous work [9, 10], all four cables were active during trajectory tacking unlike in this

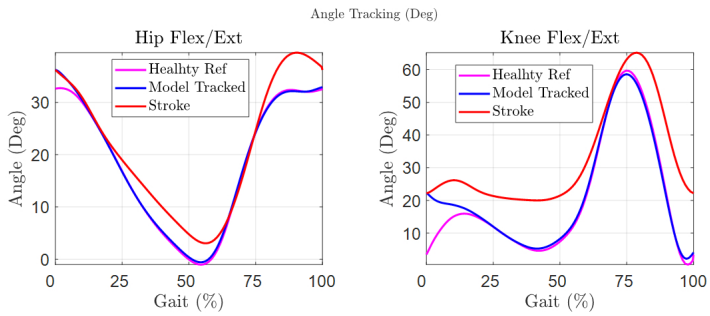

Figure 3: Joint angle tracking with stroke gait 1

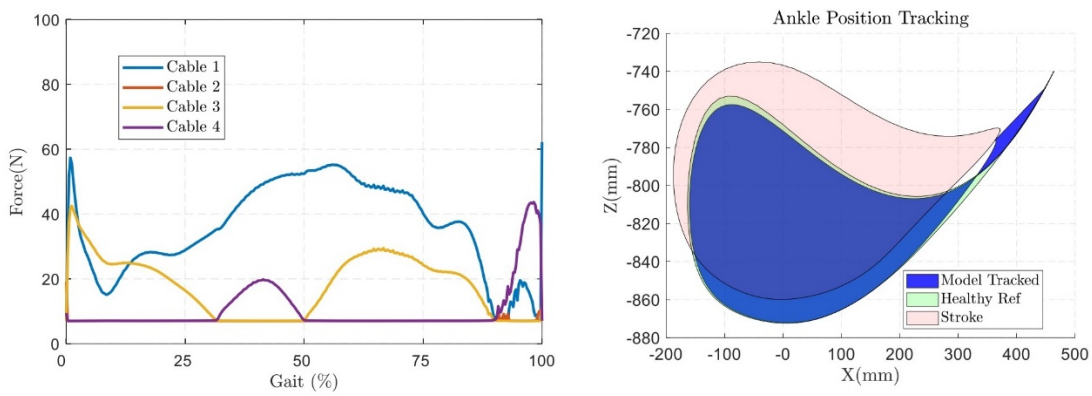

Figure 4: Required cable tension and ankle position tracking for stroke gait 1.

study, where cable 2 is not actively contributing to the impaired trajectory correction (Figure 4). This is due to the inclusion of the subject's active contribution (stroke gait contribution) during tracking along with the passive elastic joint moment.

3.2 Tracking with Stroke Gait 2

The tracking of reference trajectory with stroke gait 2 is shown in Figure 5. The stroke gait has a higher reduced knee range of motion (ROM). C-LREX tracked the reference trajectory with some deviation at the hip and knee joints. The tracked cable tension and ankle position are shown in Figure 6. Due to reduced knee ROM (flexion), the C-LREX required higher cable tension to correct the knee flexion angle (Cable 3). Similarly, the stroke leg has reduced ROM for the hip joint, thus required larger tension in the hip extension cable (Cable 1). However, due to the limited range of cable tension, the model failed to generate the required joint moment during 60-80% of the gait cycle and to fully correct the deviated trajectory.

3.3 Tracking with Stroke Gait 3

Similar to stroke gait 2, stroke gait 3 has reduced knee and hip ROM, specifically knee flexion and hip extension (Figure 7). Thus, C-LREX applied higher cable tension via cable 1 and cable 3 to meet the reference trajectory and (Figure 8). Since the knee ROM is higher compared to stroke gait 2, the required cable tension is lower and stays within the specified tension limit.

3.4 Tracking with Stroke Gait 4

The tracking of healthy trajectory with stroke gait 4 is shown in Figure 9. The impairment has reduced hip ROM and required higher cable tension in the hip cable (cable 1) to track the reference trajectory (Figure 10).

3.5 Tracking with Stroke Gait 5

The tracking performance and cable tension requirements are shown in Figure 11 and Figure 12. Stroke gait 5 has a higher reduction in the ROM of the hip joint and the knee joints. In addition, the higher inertial properties of the thigh section compared to the shank section resulted in a larger joint moment required to correct the hip's ROM and thus higher cable tension. Due to the limitation on cable tension, the model failed to track the healthy reference trajectory closely as the requirements exceeds the range during 35-70% of the gait cycle. Although the knee cable tension remained well within the specified limit, the model failed to track the knee trajectory closely due to the fact that the knee extension moment will further increase requirements in the hip flexion moment.

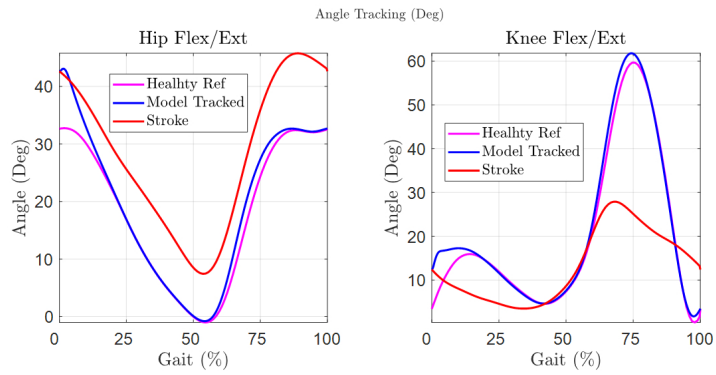

Figure 5: Joint angle tracking with stroke gait 2

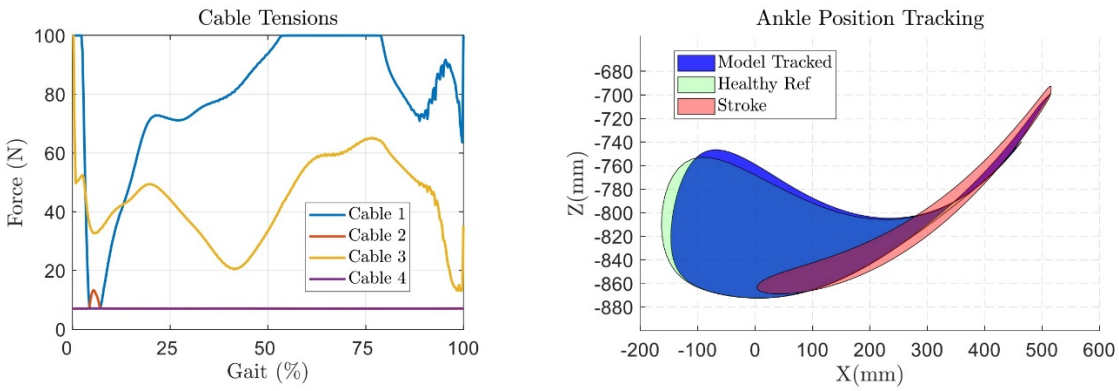

Figure 6: Applied cable tension and ankle position tracking with stroke gait 2.

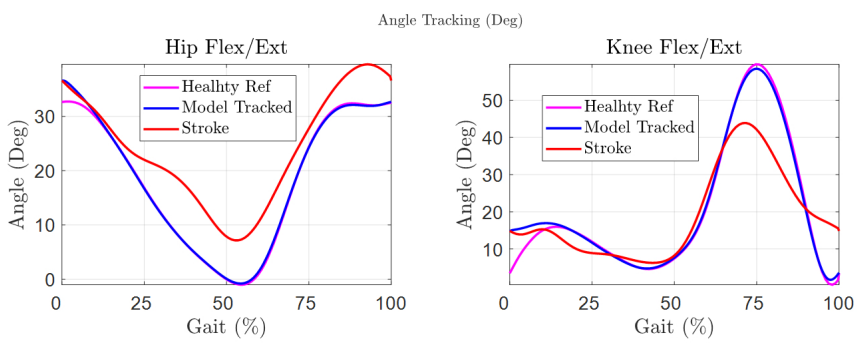

Figure 7: Joint angle tracking with stroke gait 3

The tracking errors in the ankle trajectory in all stroke gaits (1-5) are compared in Figure 13. Stroke gaits 1, 3, and 4 were successfully corrected with the least deviation due to their smaller overall deviation in ROM (specifically at the hip joint) while stroke gait 5 has the highest deviation. The force components (SX: anterior-posterior shear force in the X direction and SZ: compressive force in the Z direction) induced at the hip and knee joints in each stroke gait due to applied cable tension are plotted in Figure 14. C-LREX with stroke gait 1 tracks the healthy reference trajectory closely inducing the least joint force components.

4 DISCUSSION

Cable driven exoskeletons are preferred over link-driven exoskeletons due to numerous advantages including lighter weight and design modularity. In previous works, we proposed a generalized framework to analyze cable driven exoskeletons. We also proposed

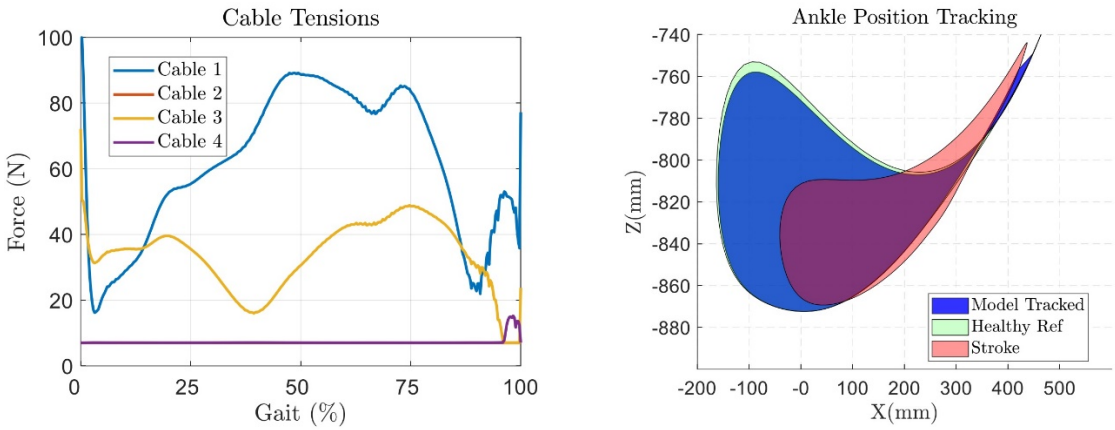

Figure 8: Applied cable tension and ankle position tracking with stroke gait 3.

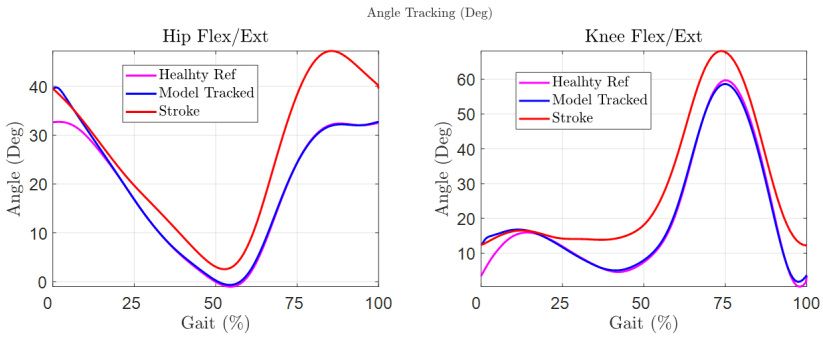

Figure 9: Joint angle tracking with stroke gait 4

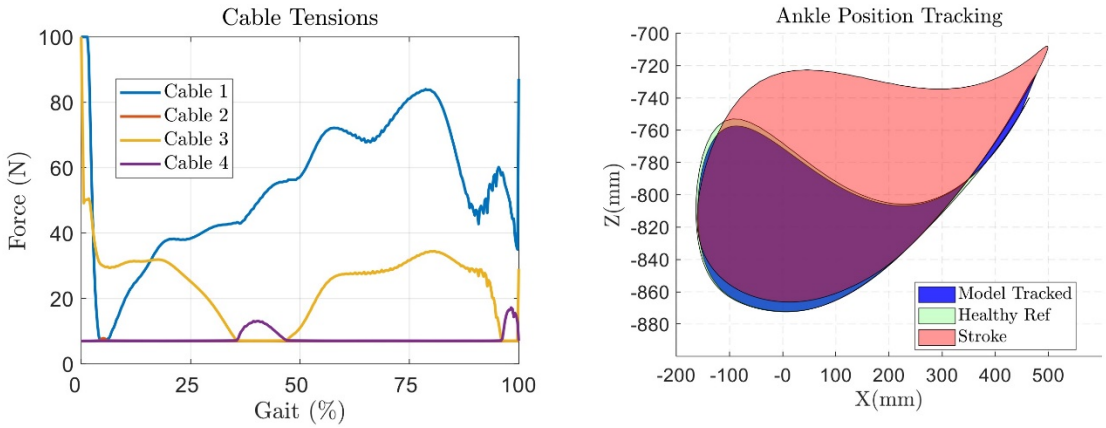

Figure 10: Applied cable tension and ankle position tracking with stroke gait 4.

a four-cable driven exoskeleton C-LREX as a potential design solution to assist impaired gait to track a reference healthy trajectory. The lower limb model considers the passive elastic joint moment, the limb's mass, and inertial properties as well as the possible partial active contribution of the impaired limb to achieve impaired kinematics. The C-LREX can assist various levels of impaired gait as long as the required cable tension falls within the specified range. Nevertheless, the generalized framework allows model modification to accommodate higher cable tension requirements. The passive elastic joint moment was calculated using the relations [14] (depends only on the joint angles) and the inertial properties were

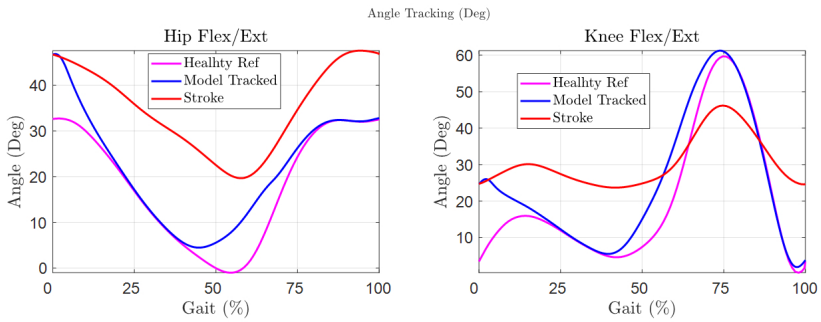

Figure 11: Joint angle tracking with stroke gait 5

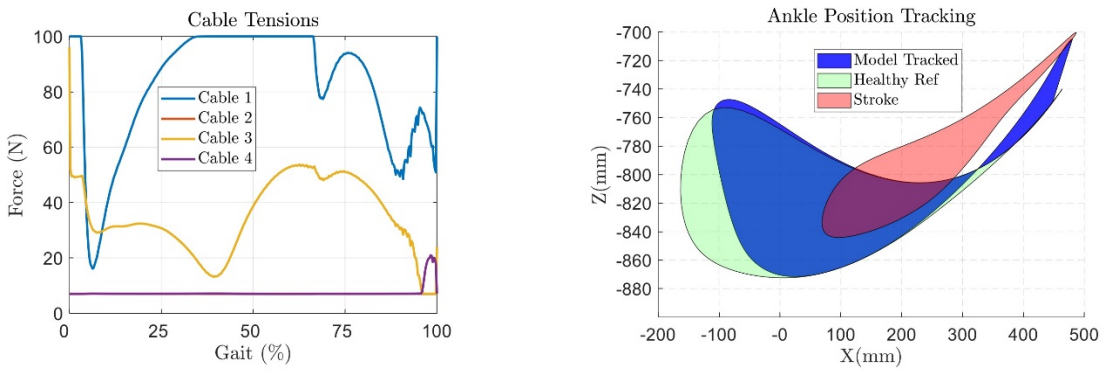

Figure 12: Applied cable tension and ankle position tracking with stroke gait 5.

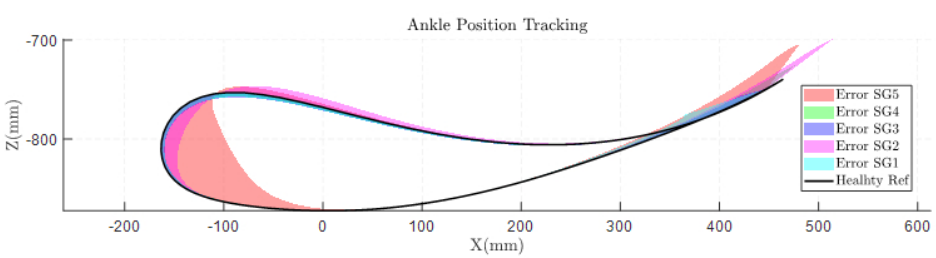

Figure 13: Error in ankle position tracking. (SG refers to Stroke Gait)

estimated using [15]. In the C-LREX model, the physiological condition of impaired gait (data on deteriorated stiffness or spasticity) was not included. However, the current framework [10] is open to considering such information provided that a quantitative model exists.

The gait of five different stroke survivors was adopted from the literature but the model used the anthropometric data of one

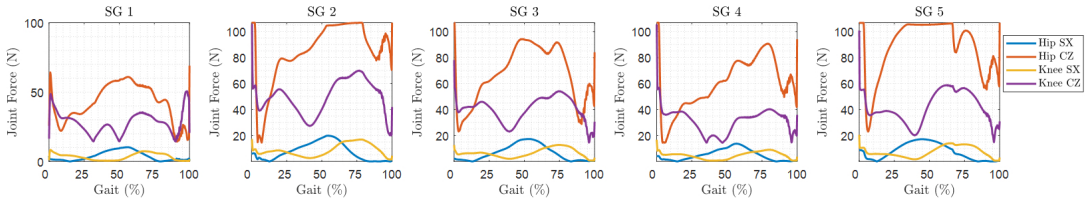

Figure 14: Joint force components induced at the hip and knee joints due to applied cable tension. (SG refers to Stroke Gait)

individual. The C-LREX was able to assist the limb model to track the healthy trajectory except when the hip ROM was remarkedly reduced which requests a cable tension that exceeds the allowable value. The thigh has larger inertial parameters and the required moment to correct its trajectory is higher as compared to the shank. Furthermore, cable 2 in most cases remained inactive throughout the gait cycle due to the inclusion of the subject's active contribution estimated via the rigid body model along with the passive elastic joint moment. This inclusion reduced the overall required joint moment to be generated by the cables which made cable 2 and cable 4 inactive. However, the actual gait contribution of a stroke patient is estimated via a simplified model without considering muscle impairment. In an actual experimentation scenario, the patient contribution may vary, therefore, it is recommended to have 4 cables with the capability of generating antagonistic moments at the joints independently as needed.

This study demonstrated the potential of employing the C-LREX to correct the stroke gait to track as close as possible a reference healthy trajectory by comparing the hip and knee joint angles tracking and the ankle position tracking. Furthermore, it also provided an estimate of the required cable tension range to be generated by the actuators to meet the reference healthy trajectory. It is worth noting that the simulation study predicted the possible behavior of stroke gait assisted by C-LREX without actual prototyping and experimental testing.

5 CONCLUSIONS AND FUTURE WORK

Cable driven exoskeletons have numerous advantages including light weight, negligible inertia and inertial vibration exertion, and easier donning on/off as compared to the traditional link-driven exoskeleton. In previous works, cable driven exoskeletons designed to track reference healthy trajectory neglected the user's active contribution during the gait cycle. In this work, we investigated the ability of a four-cable driven model of C-LREX to correct various stroke impaired gait patterns and track a reference healthy trajectory. The C-LREX model was able to successfully correct gait with hip ROM deviation up to 15 degrees throughout the gait cycle. This limitation is due to the limited range of allowable cable tension.
This model is currently limited to sagittal plane motion tracking due to the available data on stroke survivors. Furthermore, the framework assumed the hip as the origin during the motion and thus does not include any information regarding hip hiking, a common pattern of gait in stroke patients. Future work will include hip abduction/adduction of stroke survivors in addition to the sagittal plane motion.

6 FUNDING

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